Introduction to Cells

What does it mean to be living?
Living-animals, plants
Non living- stones, sand, and summer breezes

All living things are made of cells: small, membrane-enclosed units filled with a concentrated aqueous solution of chemicals and endowed with the extraordinary ability to create copies of themselves by growing and dividing in two.

Unity and diversity of Cells

Cells are not all alike; in fact, they can be wildly different

There are at least 10 million—perhaps 100 million—distinct species of living things in the world

Cells vary enormously in appearance and function

Size- Cell vary in size upto 0.1 um to 100 um
Shape- Having different shapes such as hexagonal, oval etc.

Cells are also enormously diverse in their chemical requirements and activities.
Some require oxygen to live; for others it is deadly
Some consume little more than air, sunlight, and water as their raw materials; others need a complex mixture of molecules produced by other cells.
Some Cells are factories to produce-imp compounds like hormones, starch, fat, latex, or pigments.
Some are engines-like muscle, burning fuel to do mechanical work.
Electricity generators-like the modified muscle cells in the electric eel.

There is a division of labor among cells

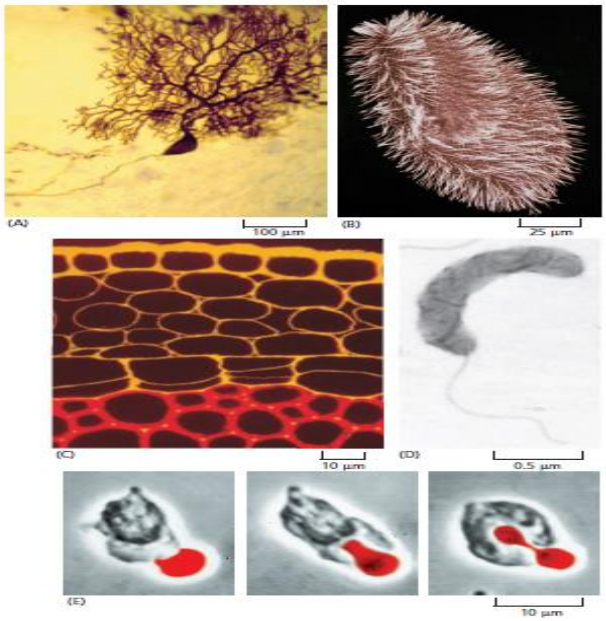


Figure 1–1 **Cells come in a variety of shapes and sizes.** (A) A nerve cell from the cerebellum (a part of the brain that controls movement). This cell has a huge branching tree of processes, through which it receives signals from as many as 100,000 other nerve cells. (B) *Paramecium*. This protozoan—a single giant cell—swims by means of the beating cilia that cover its surface. (C) A section of a young plant stem in which cellulose is stained *red* and another cell wall component, pectin, is stained orange. The outermost layer of cells is at the top of the photo. (D) A small bacterium, *Bdellovibrio bacteriovorus*, that uses a single terminal flagellum to propel itself. This bacterium attacks, kills, and feeds on other, larger bacteria. (E) A human white blood cell (a neutrophil) approaching and engulfing a red blood cell. (A, courtesy of Constantino Sotelo; B, courtesy of Anne Fleury, Michel Laurent, and André Adoutte; D, courtesy of Murry Stein; E, courtesy of Stephen E. Malawista and Anne de Boisfleury Chevance.)

Living Cells all have a similar Basic Chemistry

Despite the extraordinary diversity of plants and animals there is something common

Cells sharing the same machinery for the most basic functions.

All cells are composed of the same sorts of molecules that participate in the same types of chemical reactions

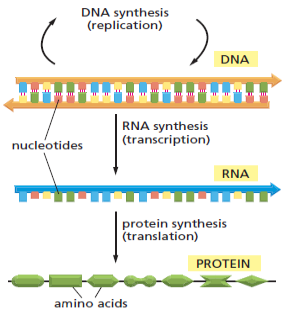


Figure 1–2 **In all living cells, genetic information flows from DNA to RNA (transcription) and from RNA to protein (translation).** Together these processes are known as gene expression.

Thus, in every cell, the long DNA polymer chains are made from the same set of four monomers, called *nucleotides*, *strung* together in different sequences like the letters of an alphabet to convey different information.

In every cell, the instructions in the DNA are read out, or *transcribed, into a chemically related set of polymers called RNA*

the messages carried by m*RNA* are in turn *translated into yet another type of polymer called a protein.*

Protein molecules dominate the behavior of the cell, serving as structural supports, chemical catalysts, molecular motors, and so on.

Viruses, for example, are compact packages of genetic information—in the form of DNA or RNA

Viruses have no ability to reproduce themselves by their own efforts.

They get themselves copied by parasitizing the reproductive machinery of the cells that they invade. Thus, viruses are chemical zombies: they are inert and inactive outside their host cells but exert a malign control once they gain entry.

All present-day Cells have apparently evolved from the same ancestor

All cells duplicates-DNA replication- daughter cells resemble the parent cell

However, the copying is not always perfect, and the instructions are occasionally corrupted by *mutations that change the DNA.*

That is why daughter cells do not always match the parent cell exactly.

Mutations in offspring that are changed for the worse (less able to survive and reproduce)
changed for the better (in that they are better able to survive and reproduce)
changed neutrally (in that they are genetically different but equally viable)
The struggle for survival eliminates the first, favors the second, and tolerates the third.

Evolution—the process by which living species become gradually modified and adapted to their environment in more and more sophisticated ways.

Evolution offers a startling but compelling explanation of why present-day cells are so similar in their fundamentals: they have all inherited their genetic instructions from the same common ancestor.

It is estimated that this ancestral cell existed between 3.5 billion and 3.8 billion years ago, and we must suppose that it contained a prototype of the universal machinery of all life on Earth today.

Genes provide the instructions for Cellular form, function, and Complex Behavior

A cell's genome- that is, the entire library of genetic information in its DNA

Provides a genetic program that instructs the cell how to function, and, for plant and animal cells, how to grow into an organism with hundreds of different cell types.

Cells Under the MICROSCOPE

The earliest cell biologists began by simply looking at tissues and cells, then breaking them open or slicing them up and attempting to view their contents.

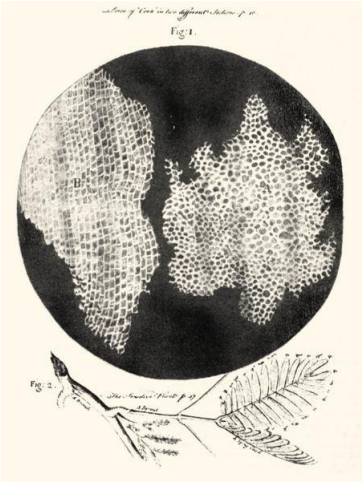
Cells were not made visible until the seventeenth century until discovery of *Light microscopes*

Electron microscopes, invented in the 1930s, go beyond this limit by using beams of electrons instead of beams of light as the source of illumination.

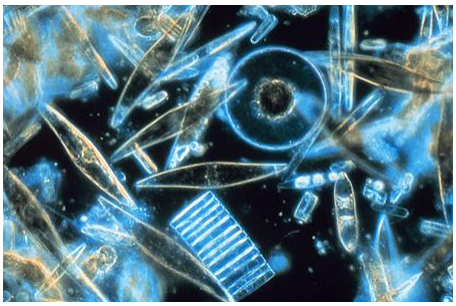
Robert Hooke examined a piece of cork and in 1665 reported to the Royal Society of London that the cork was composed of a mass of minute chambers, which he called "cells."

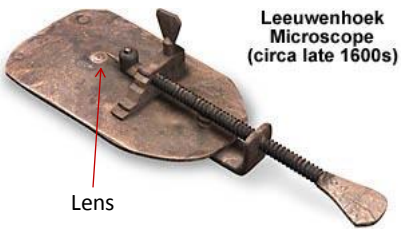
Scientists to Remember

- **Robert Hooke** (1665) – Observed "cells" in cork



- **Anton van Leeuwenhoek** (1674)
  - Father of Microscopy
  - Saw tiny living things in pond water.





Scientists

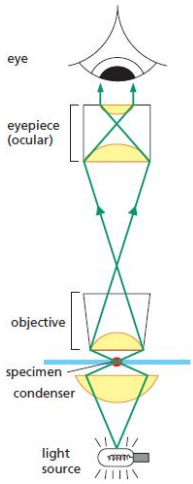
- **Robert Hooke** (1665) – Observed “cells” in cork
- **Anton van Leeuwenhoek** (1674) – Saw tiny living things in pond water.
- **Matthias Schleiden** (1838) – Plants are made of cells.
- **Theodor Schwann** (1839) – Animals are made of cells.
- **Rudolf Virchow** (1855) – New cells come from existing cells.

Scientists

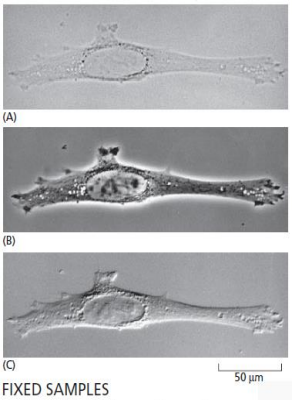
- **Janet Plowe** (1931) – Cell membrane is a physical structure.
- **Lynn Margulis** (1970) – Organelles were once free-living cells.

THE LIGHT MICROSCOPE

The light microscope allows us to magnify cells up to 1000 times and to resolve details as small as 0.2 µm (a limitation imposed by the wavelike nature of light, not by the quality of the lenses). Three things are required for viewing cells in a light microscope. First, a bright light must be focused onto the specimen by lenses in the condenser. Second, the specimen must be carefully prepared to allow light to pass through it. Third, an appropriate set of lenses (objective and eyepiece) must be arranged to focus an image of the specimen in the eye.



the light path in a light microscope

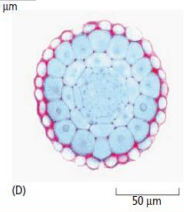


LOOKING AT LIVING CELLS

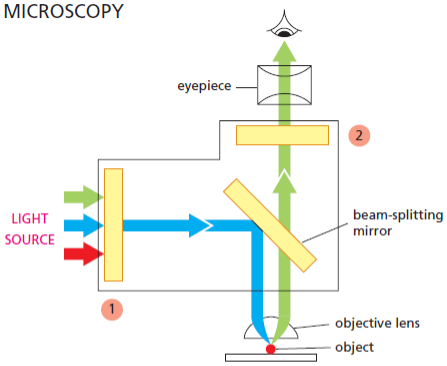
The same unstained, living animal cell (fibroblast) in culture viewed with (A) straightforward (bright-field) optics; (B) phase-contrast optics; (C) interference-contrast optics. The two latter systems exploit differences in the way light travels through regions of the cell with differing refractive indexes. All three images can be obtained on the same microscope simply by interchanging optical components.

FIXED SAMPLES

Most tissues are neither small enough nor transparent enough to examine directly in the microscope. Typically, therefore, they are chemically fixed and cut into very thin slices, or *sections*, that can be mounted on a glass microscope slide and subsequently stained to reveal different components of the cells. A stained section of a plant root tip is shown here (D). (Courtesy of Catherine Kidner.)



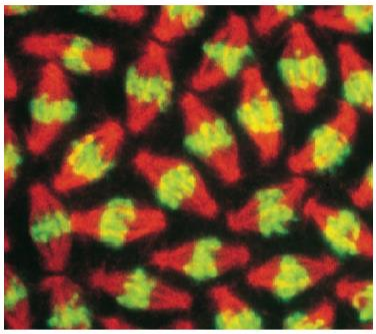
FLUORESCENCE MICROSCOPY



Fluorescent dyes used for staining cells are detected with the aid of a *fluorescence microscope*. This is similar to an ordinary light microscope except that the illuminating light is passed through two sets of filters. The first (1) filters the light before it reaches the specimen, passing only those wavelengths that excite the particular fluorescent dye. The second (2) blocks out this light and passes only those wavelengths emitted when the dye fluoresces. Dyed objects show up in bright color on a dark background.

FLUORESCENT PROBES

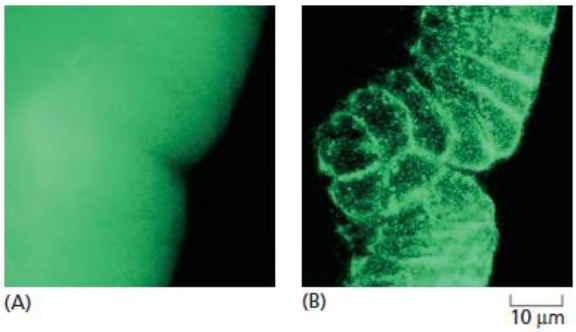
Dividing nuclei in a fly embryo seen with a fluorescence microscope after staining with specific fluorescent dyes.



Fluorescent dyes absorb light at one wavelength and emit it at another, longer wavelength. Some such dyes bind specifically to particular molecules in cells and can reveal their location when examined with a fluorescence microscope. An example is the stain for DNA shown here (*green*). Other dyes can be coupled to antibody molecules, which then serve as highly specific and versatile staining reagents that bind selectively to particular macromolecules, allowing us to see their distribution in the cell. In the example shown, a microtubule protein in the mitotic spindle is stained *red* with a fluorescent antibody. (Courtesy of William Sullivan.)

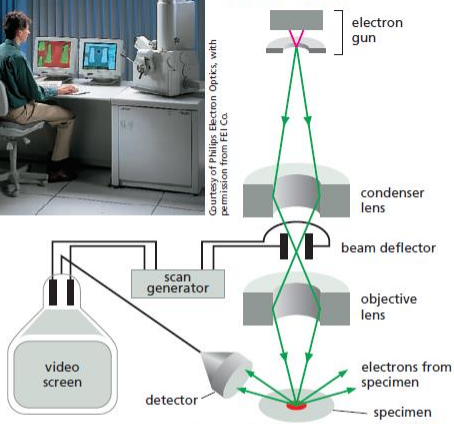
CONFOCAL MICROSCOPY

A confocal microscope is a specialized type of fluorescence microscope that builds up an image by scanning the specimen with a laser beam. The beam is focused onto a single point at a specific depth in the specimen, and a pinhole aperture in the detector allows only fluorescence emitted from this same point to be included in the image. Scanning the beam across the specimen generates a sharp image of the plane of focus—an *optical* section. A series of optical sections at different depths allows a three-dimensional image to be constructed. An intact insect embryo is shown here stained with a fluorescent probe for actin (a filamentous protein). (A) Conventional fluorescence microscopy gives a blurry image due to the presence of fluorescent structures above and below the plane of focus. (B) Confocal microscopy provides an optical section showing the individual cells clearly. (Courtesy of Richard Warn and Peter Shaw.)

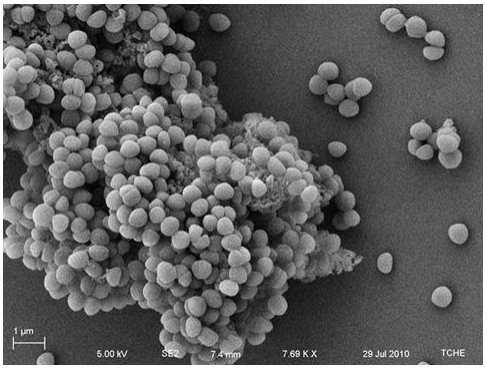


Courtesy of Philips Electron Optics, with permission from FEI Co.

SCANNING ELECTRON MICROSCOPY

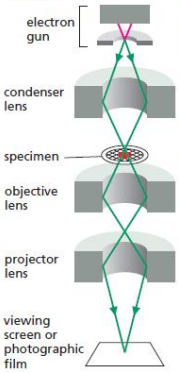


In the scanning electron microscope (SEM) the specimen, which has been coated with a very thin film of a heavy metal, is scanned by a beam of electrons brought to a focus on the specimen by the electromagnetic coils that, in electron microscopes, act as lenses. The quantity of electrons scattered or emitted as the beam bombards each successive point on the surface of the specimen is measured by the detector, and is used to control the intensity of successive points in an image built up on a video screen. The microscope creates striking images of three-dimensional objects with great depth of focus and can resolve details down to somewhere between 3 nm and 20 nm, depending on the instrument.



SEM of Staphylococcus aureus

TRANSMISSION ELECTRON MICROSCOPY



Courtesy of Philips Electron Optics, with permission from FEI Co.

The electron micrograph below shows a small region of a cell in a piece of testis. The tissue has been chemically fixed, embedded in plastic, and cut into very thin sections that have then been stained with salts of uranium and lead. (Courtesy of Daniel S. Friend.)

The transmission electron microscope (TEM) is in principle similar to a light microscope, but it uses a beam of electrons instead of a beam of light, and magnetic coils to focus the beam instead of glass lenses. The specimen, which is placed in a vacuum, must be very thin. Contrast is usually introduced by staining the specimen with electron-dense heavy-metals that locally absorb or scatter electrons, removing them from the beam as it passes through the specimen. The TEM has a useful magnification of up to a million-fold and with biological specimens can resolve details as small as about 2 nm.

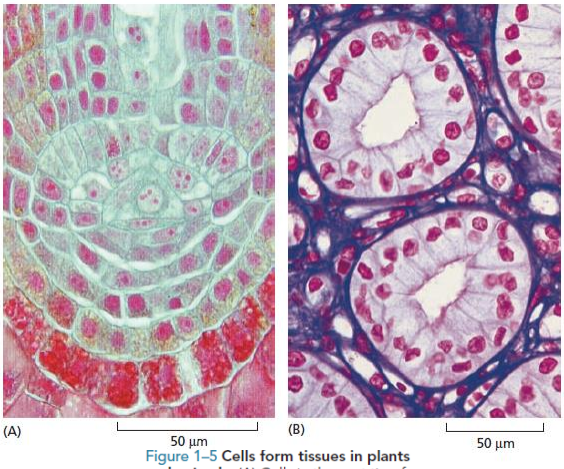
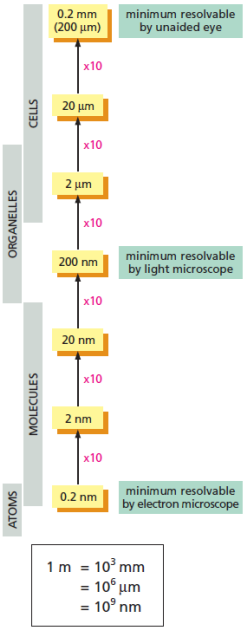


Figure 1–5 **Cells form tissues in plants and animals.** (A) Cells in the root tip of a fern. The nuclei are stained *red* and each cell is surrounded by a thin cell wall *(blue)*. (B) Cells in the urine-collecting ducts of the kidney. Each duct appears in this cross section as a ring of closely packed cells (with nuclei stained *red*). The ring is surrounded by extracellular matrix, stained *purple*. (A, courtesy of James Mauseth; B, from P.R. Wheater et al., Functional Histology, 2nd ed. Edinburgh: Churchill Livingstone, 1987. With permission from Elsevier.)



To see the internal structure of a cell is difficult, not only because the parts are small but also because they are transparent and mostly colorless.

One approach is to stain cells with dyes that color particular components differently

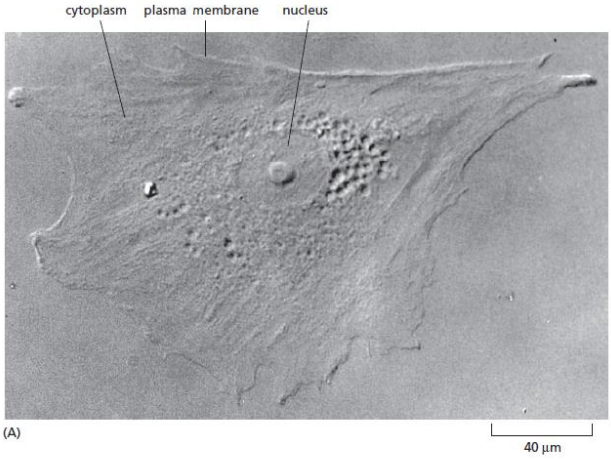
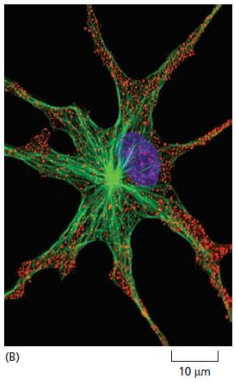


Figure 1–7 **The internal structures of a living cell can be seen with a light microscope.** (A) A cell taken from human skin and grown in tissue culture was photographed through a light microscope using interference-contrast optics (see Panel 1–1, pp. 8–9). The nucleus is especially prominent. (B) A pigment cell from a frog, stained with fluorescent dyes and viewed with a confocal microscope (see Panel 1–1). The nucleus is shown in *blue*, the pigment granules in *red*, and the microtubules—a class of filaments built from protein molecules in the cytoplasm—in *green*. (A, courtesy of Casey Cunningham; B, courtesy of Steve Rogers and the Imaging Technology Group.)



When thin sections are cut, stained, and placed in the electron microscope, much of the jumble of cell components becomes sharply resolved into distinct *organelles—separate, recognizable substructures that are* only hazily defined under the light microscope.

Cell Theory

- Confirmed discoveries that all scientists believe to be true about cells:
  - 1. Cells are the basic unit of life.
  - 2. All living things are made of cells.
  - 3. New cells are produced from existing cells.

Unicellular Cell

Made up of single cell

Has top perform all the functions such as reproduction, growth and metabolism, protection etc.

To perform all these functions cell has different cell organelles

For protection cell has rigid cell wall which gives shape and protection

Unicellular Cell has rapid growth rate such as bacterial cell divide each 20 min

Disadvantages

1) No division of labour- Single cell has to perform all the functions

2) Cell is exposed to extracellular environment so its difficult to protect cell

3) Cell has to sufficiently large to accommodate all the cell organelles so increase in size decreases surface to volume ration and also large size is difficult to control for by nucleus of exchange of material and information from the cell surface

4) Death of cell kill the organism

5) Life span is short and involves disappearance of parent and formation of reproductive structure

Multicellular Cell

Made up of more than one cell

Advantages

1) Dual function- For cell and organism
2) Reduction in workload
3) Increased efficiency- Different specialization of the cells for a particular function hence efficiency is high eg Nerve cell
4) Size and shape- Cells are usually small
5) Constant internal environment- The interstitial space having interstitial fluid so environment is constant
6) Surface cell (Protection from external environment) eg. Cork cells, Skin cells, Feathers, Nails
7) Division of labours
8) Loss of basic activities – Even though loss of function Some dead cell may have imp function eg. Skin cell

9) Cell replacement – Death of cell does not kill the organism
10) Higher survival

Disadvantages

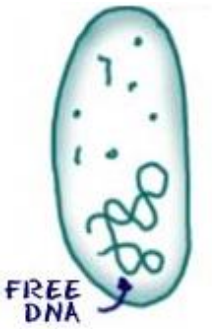
1) Specialized cell often lose power of regeneration so injury is not repaired eg Nerve cells
2) Regeneration ability of cell decreases with the Specialization
3) Specialized cell may lose vital function eg RBC
4) Unicellular cell are immortal by dividing but in multicellular cell immortality is restricted to small number of cell eg somatic cell

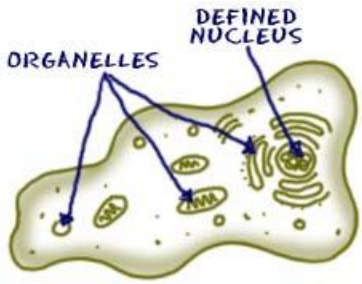
The Discovery of Cells

Prokaryotic

before

nucleus



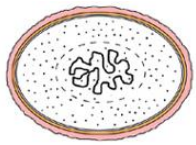


Eukaryotic

true

nucleus

Two Basic Cell Types: Prokaryotic vs. Eukaryotic Cells



Prokaryotic Cell

Animal (Eukaryotic) Cell

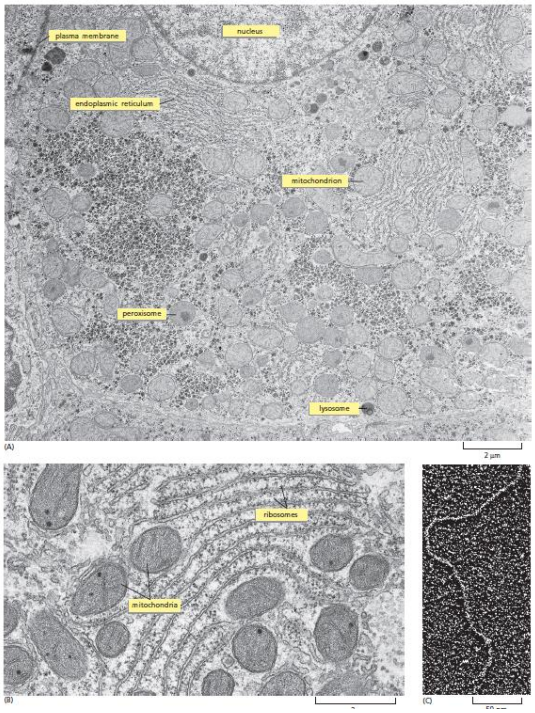
Two Basic Types

- Remember….cells are the basic unit of life for ALL living things.

- There are two basic types of cells:
  1. Prokaryotic cells – found in bacteria
  2. Eukaryotic cells – found in protists, fungi, plants and animals

Characteristics Shared

- Perform the same basic functions
- Surrounded by plasma membrane to control what enters and leaves the cell
- “Filled” with cytoplasm
- Contain ribosomes to make protein
- Contain DNA to give the general instructions for the cell’s life



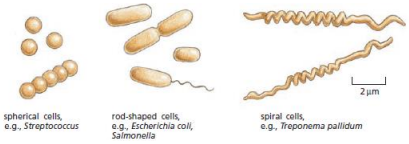


Figure 1–10 **Bacteria come in different shapes and sizes.** Typical spherical, rodlike, and spiral-shaped bacteria are drawn to scale. The spiral cells shown are the organisms that cause syphilis.

Characteristics of prokaryotic cell

Prokaryotes are typically spherical, rodlike, or corkscrew-shaped, and small—just a few micrometers long.

They often have a tough protective coat, called a cell wall, surrounding the plasma membrane, which encloses a single compartment containing the cytoplasm and the DNA.

The cells reproduce quickly by dividing in two. Under optimum conditions, when food is plentiful, a prokaryotic cell can duplicate itself in as little as 20 minutes. In 11 hours, by repeated divisions, a single prokaryotic can give rise to more than 8 billion progeny (which exceeds the total number of humans presently on Earth).

Thanks to their large numbers, rapid growth rates, and ability to exchange bits of genetic material by a process akin to sex, populations of prokaryotic cells can evolve fast, rapidly acquiring the ability to use a new food source or to resist being killed by a new antibiotic.

What Makes Eukaryotic Cells Different?

Eukaryote = with a nucleus

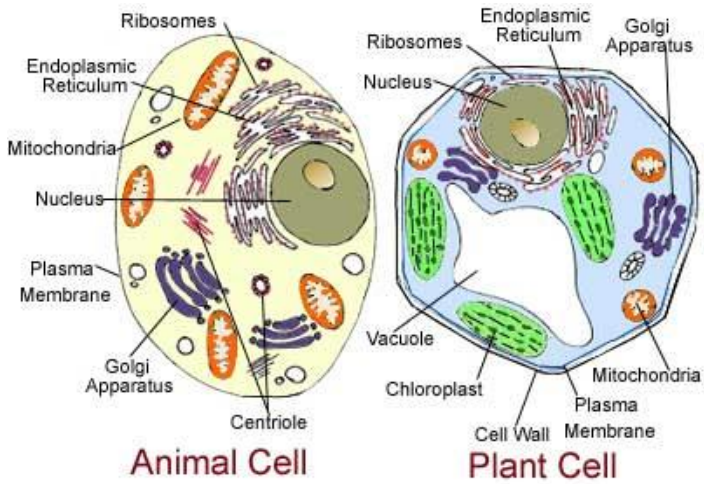
- Much larger
- Much more complex

- Contain a true nucleus to house the genetic material (DNA)
- Linear DNA packaged into chromatin found inside the nucleus
- Contains specialized structures in the cytoplasm called organelles to carry out various functions
- Not all have a cell wall

Components of Eukaryotic cell

- Cytoplasm
- Nucleus
- Mitochondria
- Chloroplast
- Ribosomes
- Golgi body
- Vacuoles
- Lysosomes
- Cytoskeleton
- Centriole
- Cilium and Flagellum
- Microvilli
- Cell membrane
- Cell Wall

Eukaryotic Cells



AN ANIMAL CELL

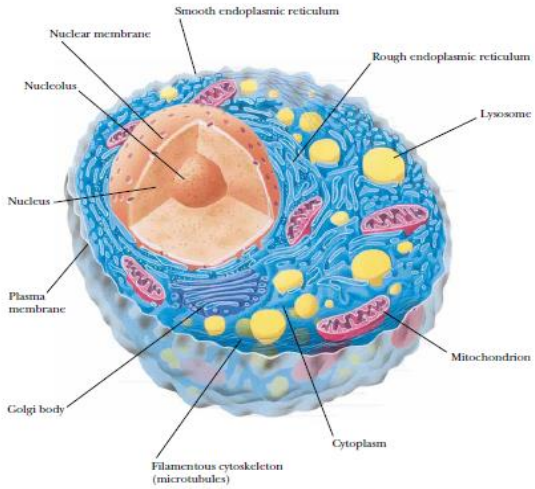


FIGURE 1.22 • This figure diagrams a rat liver cell, a typical higher animal cell in which the characteristic features of animal cells are evident, such as a nucleus, nucleolus, mitochondria, Golgi bodies, lysosomes, and endoplasmic reticulum (ER). Microtubules and the network of filaments constituting the cytoskeleton are also depicted. (*photos, top, Dwight R. Kuhn/Visuals Unlimited; middle, D.W. Fawcett/Visuals Unlimited; bottom, Keith Porter/Photo Researchers, Inc.*)

A PLANT CELL

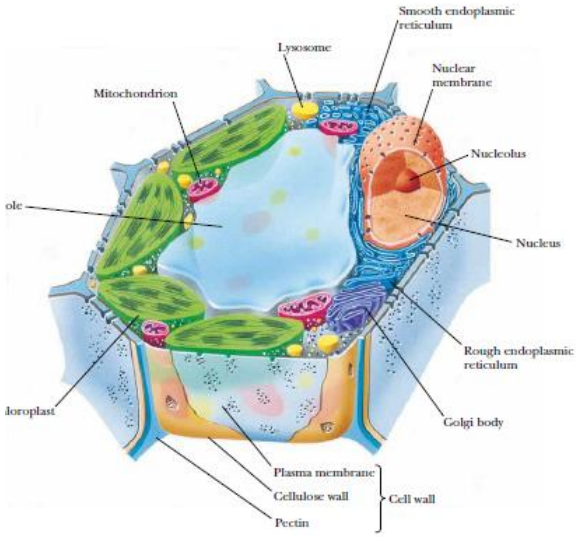


FIGURE 1.23 • This figure diagrams a cell in the leaf of a higher plant. The cell wall, membrane, nucleus, chloroplasts, mitochondria, vacuole, ER, and other characteristic features are shown. (*photos, top, middle, Dr. Dennis Kunkel/Phototake, NYC; bottom, Biophoto Associates*)